

Shih-Ni Prim  
[sprim@ncsu.edu](mailto:sprim@ncsu.edu)  
(336)254-3431

April 2, 2025

Institutional Strategy and Analysis (ISA)  
North Carolina State University  
Raleigh, NC 27607

Dear hiring committee,

I am writing to express my interest in the position of data scientist at Institutional Strategy and Analysis (ISA). Currently a third-year PhD student in statistics at North Carolina State University, I am experienced with developing novel statistical and machine learning methods, programming with R and Python, conducting exploratory data analysis, creating visualization to understand data and to communicate findings, and presenting my research in written and verbal formats. With my experience as a research assistant at NC State, summer intern at Lawrence Livermore National Laboratory, and Oak Ridge Institute of Science and Education (ORISE) fellow at US EPA, I believe I can do more than entry level work and decided to apply for both the data scientist (entry level) and data scientist positions.

As a research assistant, I worked with a biological and agricultural engineering professor and an algal physiology and ecology professor to analyze time series data to find factors that contribute to harmful algal bloom. The project required extensive data cleaning and preprocessing, data analysis and modeling, visualization, and writing of journal publication. During my time in the master's and PhD program, I have worked in a wide variety of research projects. The research projects for my courses include tensor decomposition, using machine learning methods to predict critical temperatures for semiconductors, causal inference for multiple exposures and multiple outcomes, transfer learning, etc. I have also helped with writing an online pamphlet for helping natural resource management specialists understand machine learning methods. I believe my experience in various data science projects and in collaborating with researchers outside statistics has prepared me for becoming a data scientist.

Besides assisting with others' research, my own doctoral research requires creativity and critical thinking to solve problems. My research includes two very different projects. One project develops a fully Bayesian, spectral method to analyze the effects of multiple exposures with the application of analyzing health burdens of per- and polyfluoroalkyl substances (PFAS). A second project designs an active learning method to optimize how to best sample points for a surrogate model (using deep Gaussian Process) for computer experiments with the application of finding best conditions for stable and efficient flights. The wild differences between the two projects in terms of data and collaborators—geospatial data and epidemiologists as collaborators versus

fluid dynamic simulator data and aerospace engineers as collaborators—have helped me gain breadth in knowledge about machine learning modeling. For both projects, my advisor, mentors, and I develop novel statistical and machine learning methods, and I have learned how to work independently and collaborate with my mentors to solve problems at different stages. I have also gained experience in written and verbal communication skills as well as creating visualizations through writing the manuscripts and presenting these projects at conferences.

I hope I have clearly outlined my qualifications and deep interest in this position. If I were to be selected for this role, I plan to continue working on my doctoral dissertation outside work time. Since I am a student at NC State (Go Pack!), I believe this role will help me gain work experience before graduation while allowing me to finish my degree. Thank you for your time and consideration. I am looking forward to hearing from you.

Sincerely,

Shih-Ni Prim